Name : Dudakiya Rudrax Hasmukhbhai

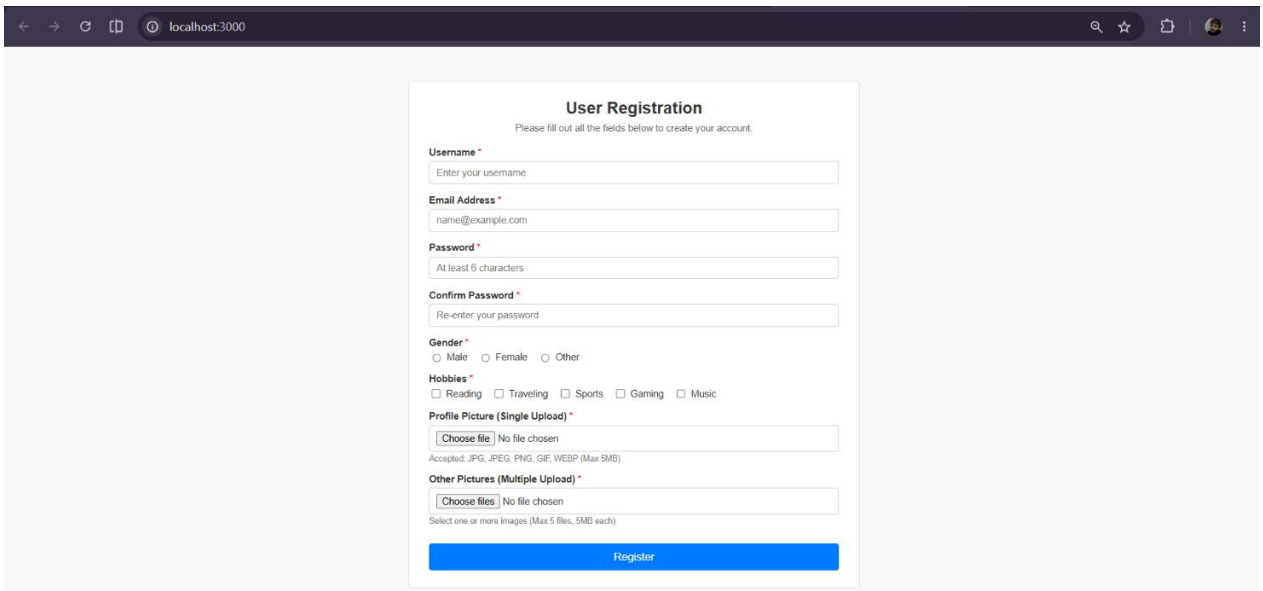
Roll No : 11

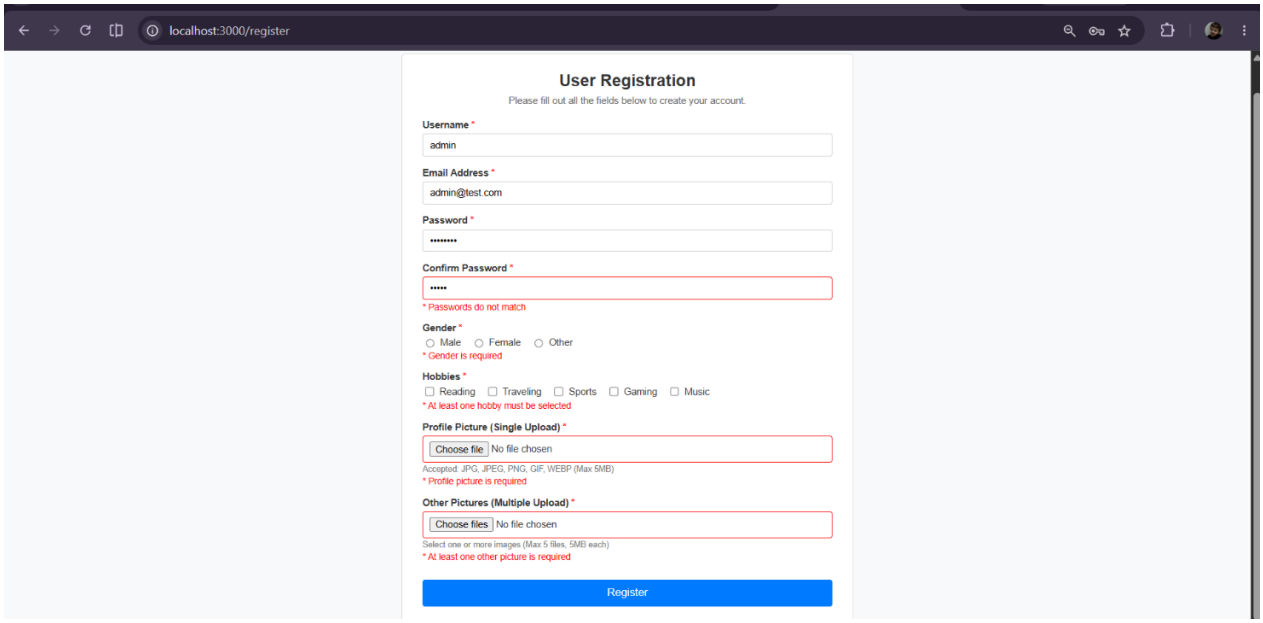
Div : A

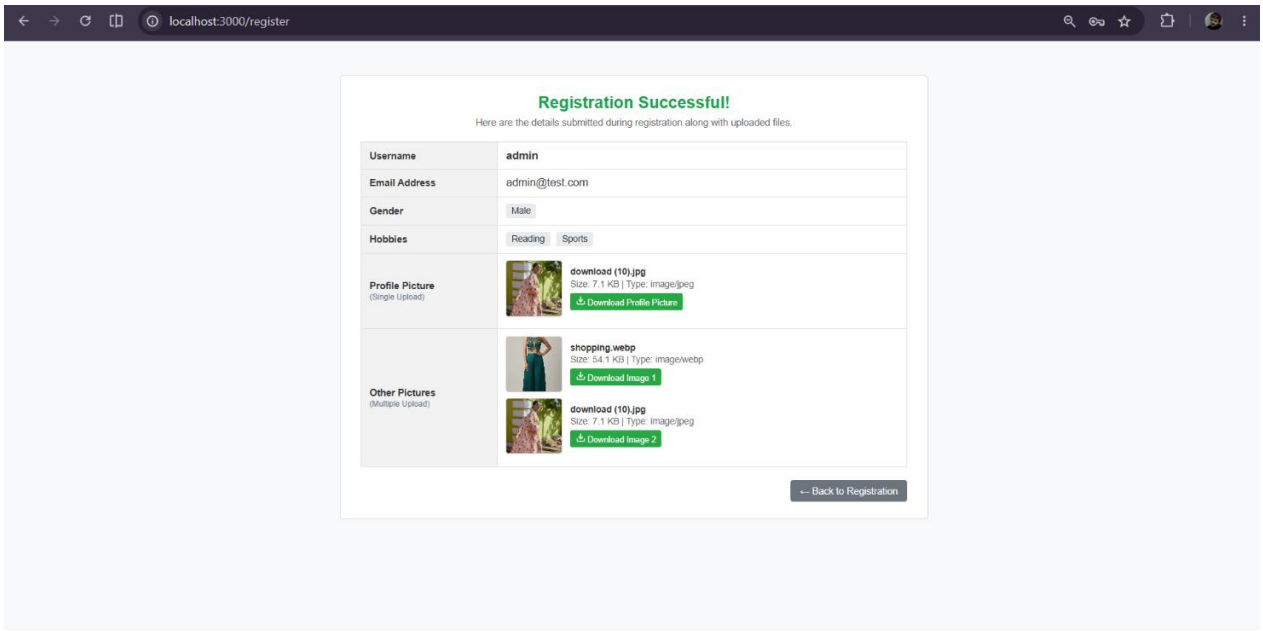
Subject : 701 Fullstack Develeopment

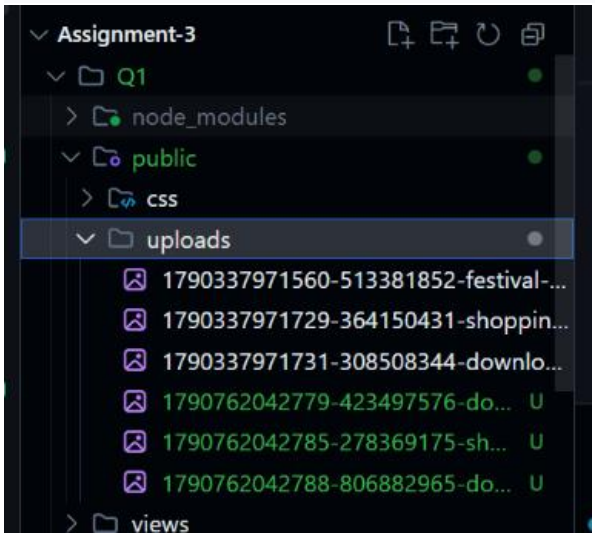
Output Pdf

Q1

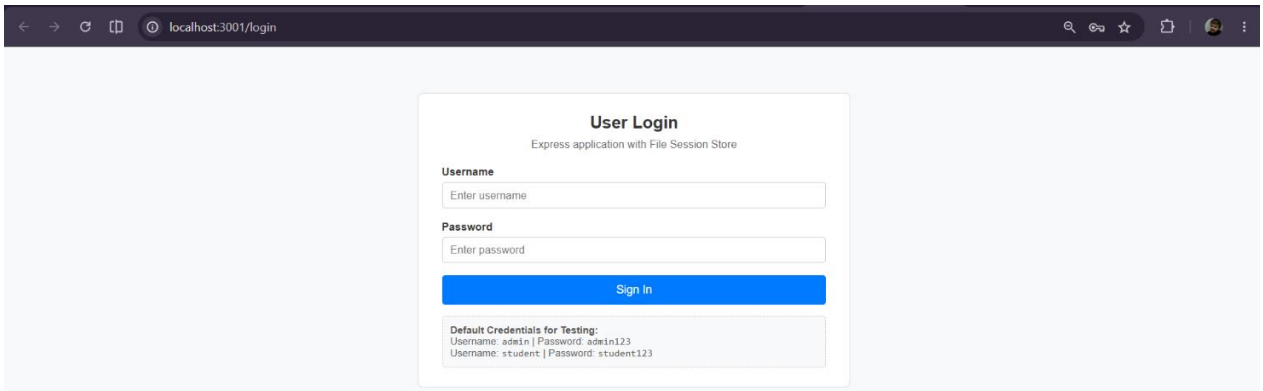

localhost:3000
User Registration
Please fill out all the fields below to create your account.
Username *
Enter your username
Email Address *
name@example.com
Password *
At least 6 characters
Confirm Password *
Re-enter your password
Gender *
Male Female Other
Hobbies *
Reading Traveling Sports Gaming Music
Profile Picture (Single Upload) *
Choose file No file chosen
Accepted: JPG, JPEG, PNG, GIF, WEBP (Max 5MB)
Other Pictures (Multiple Upload) *
Choose files No file chosen
Select one or more images (Max 5 files, 5MB each)
Register

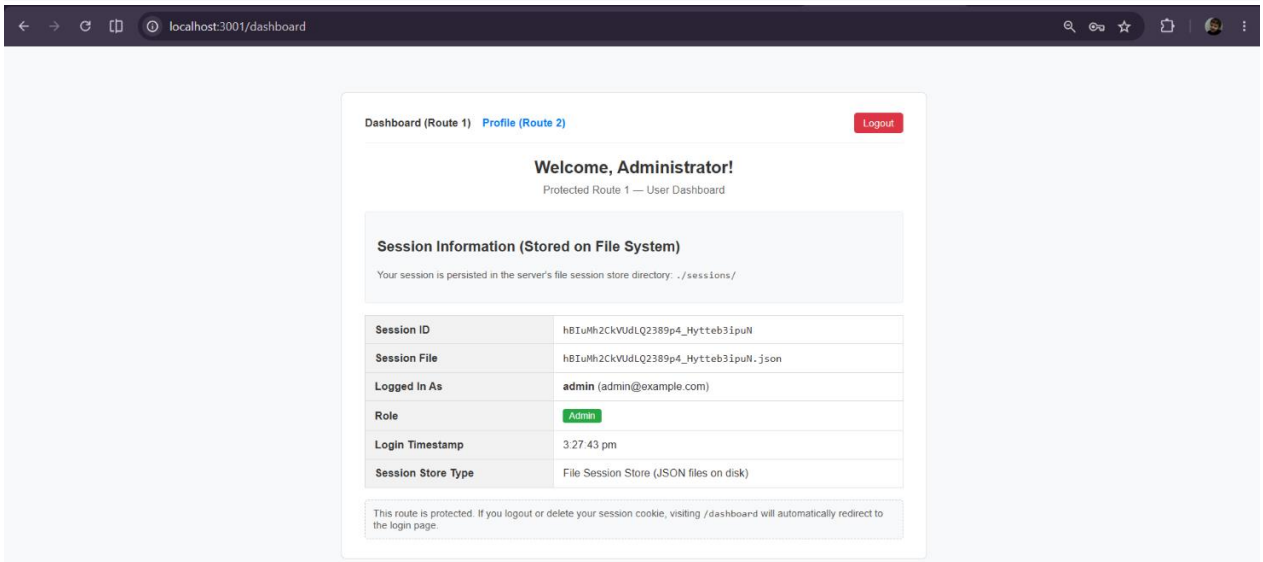

localhost:3000/register
User Registration
Please fill out all the fields below to create your account.
Username *
admin
Email Address *
admin@test.com
Password *
Confirm Password *
* Passwords do not match
Gender *
Male Female Other
* Gender is required
Hobbies *
Reading Traveling Sports Gaming Music
* At least one hobby must be selected
Profile Picture (Single Upload) *
Choose file No file chosen
Accepted: JPG, JPEG, PNG, GIF, WEBP (Max 5MB)
* Profile picture is required
Other Pictures (Multiple Upload) *
Choose files No file chosen
Select one or more images (Max 5 files, 5MB each)
* At least one other picture is required
Register

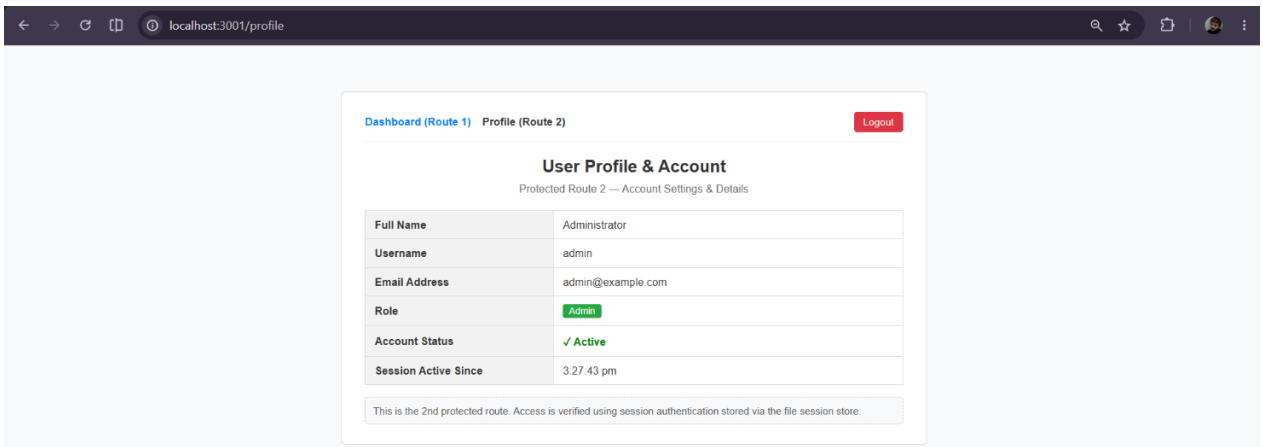

localhost:3000/register
Registration Successful!
Here are the details submitted during registration along with uploaded files.
Username
admin
Email Address
admin@test.com
Gender
Male
Hobbies
Reading
Sports
Profile Picture
(Single Upload)
download (10).jpg
Size: 7.1 KB | Type: image/jpeg
Download Profile Picture
Other Pictures
(Multiple Upload)
shopping.webp
Size: 54.1 KB | Type: image/webp
Download Image 1
download (10).jpg
Size: 7.1 KB | Type: image/jpeg
Download Image 2
← Back to Registration

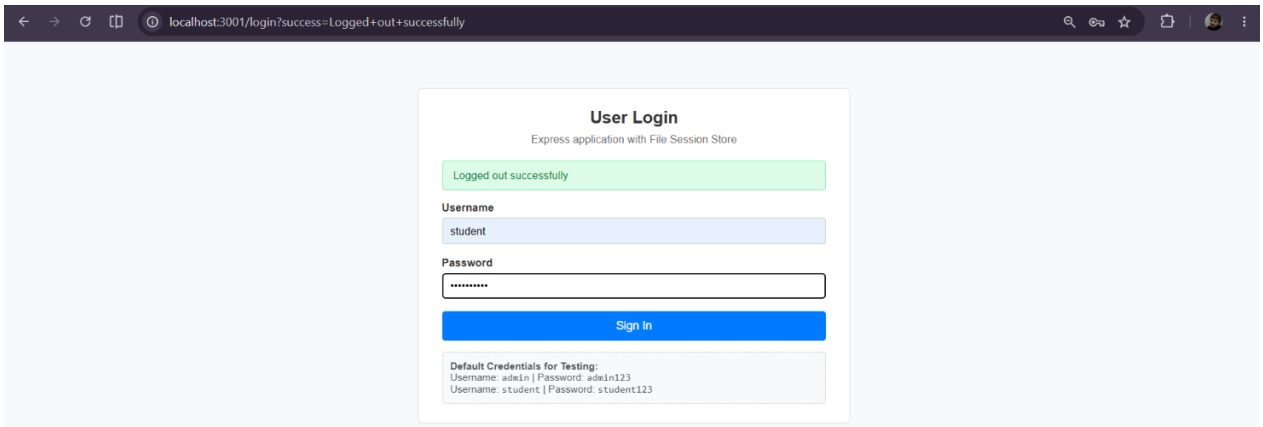

Assignment-3
Q1
node_modules
public
css
uploads
1790337971560-513381852-festival-...
1790337971729-364150431-shoppin...
1790337971731-308508344-downlo...
1790762042779-423497576-do... U
1790762042785-278369175-sh... U
1790762042788-806882965-do... U
views

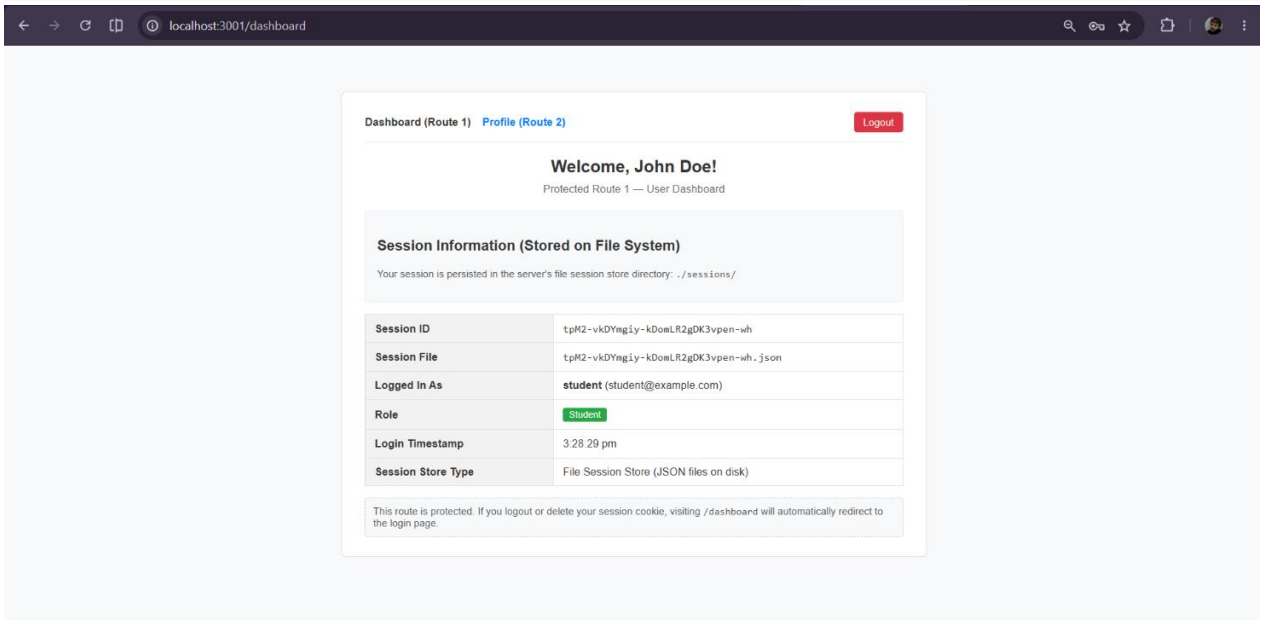
Q2

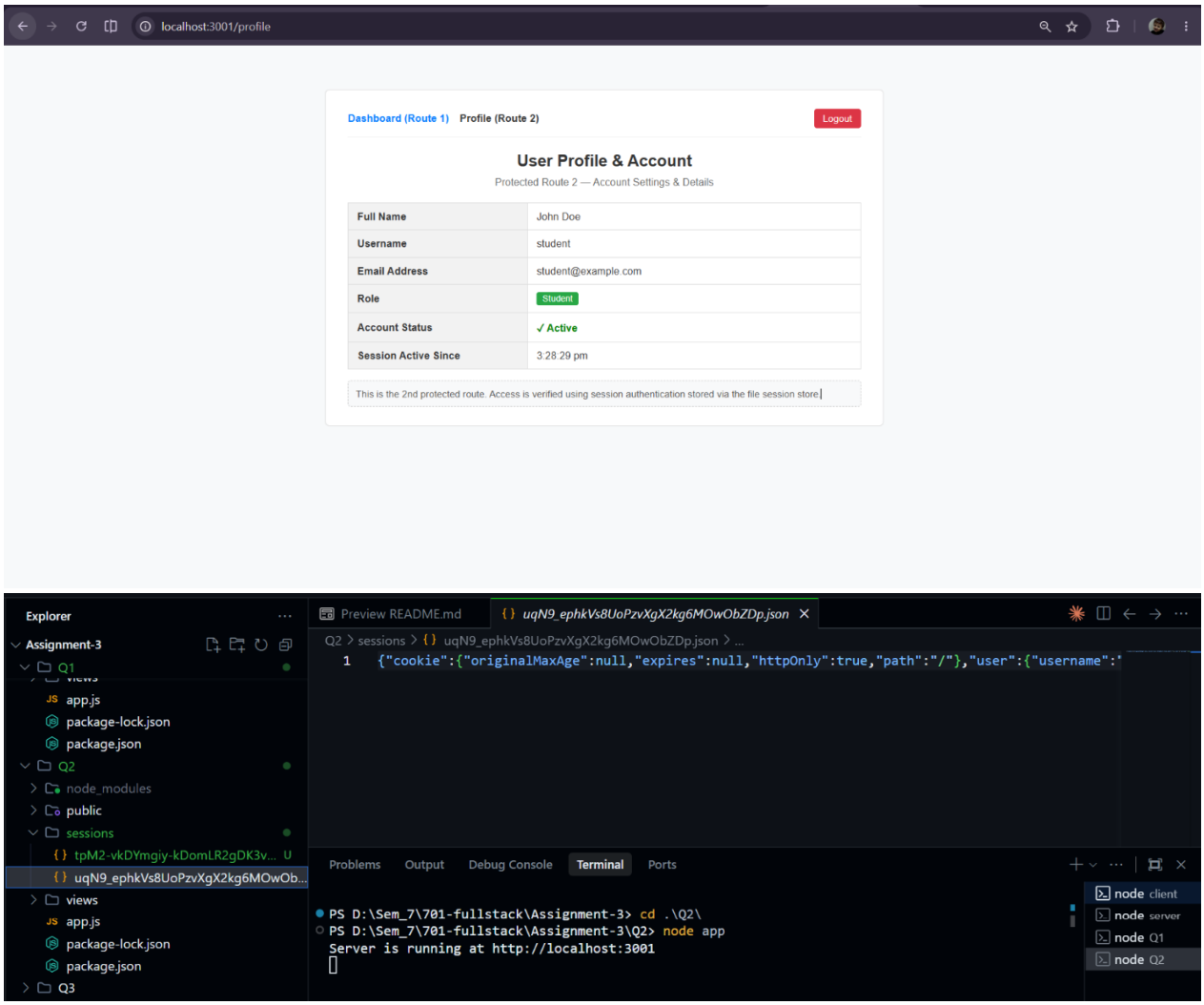

localhost:3001/login
User Login
Express application with File Session Store
Username
Enter username
Password
Enter password
Sign In
Default Credentials for Testing:
Username: admin | Password: admin123
Username: student | Password: student123


localhost:3001/dashboard
Dashboard (Route 1)
Profile (Route 2)
Logout
Welcome, Administrator!
Protected Route 1 — User Dashboard
Session Information (Stored on File System)
Your session is persisted in the server's file session store directory: ./sessions/
Session ID
hBIuMh2CkVUdlQ2389p4_Hytteb3ipuN
Session File
hBIuMh2CkVUdlQ2389p4_Hytteb3ipuN.json
Logged In As
admin (admin@example.com)
Role
Admin
Login Timestamp
3:27:43 pm
Session Store Type
File Session Store (JSON files on disk)
This route is protected. If you logout or delete your session cookie, visiting /dashboard will automatically redirect to the login page.

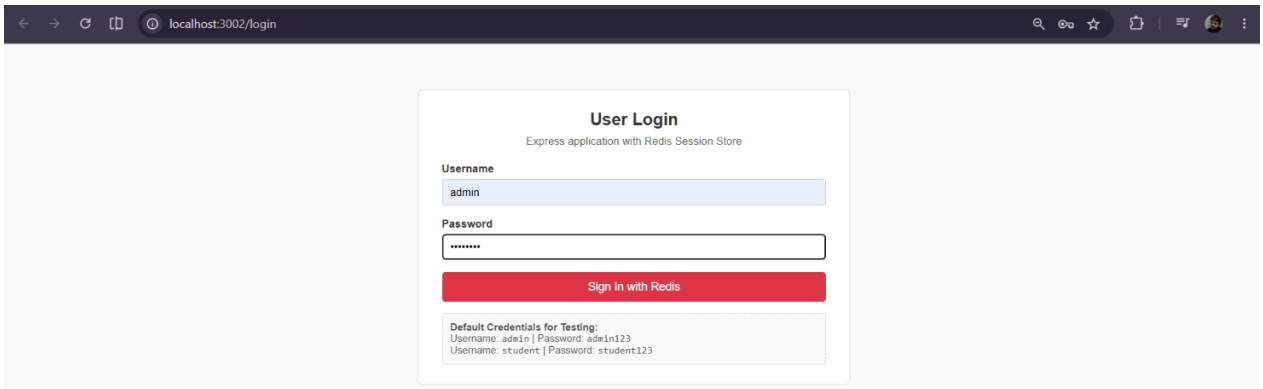

localhost:3001/profile
Dashboard (Route 1)
Profile (Route 2)
Logout
User Profile & Account
Protected Route 2 — Account Settings & Details
Full Name
Administrator
Username
admin
Email Address
admin@example.com
Role
Admin
Account Status
✓ Active
Session Active Since
3:27:43 pm
This is the 2nd protected route. Access is verified using session authentication stored via the file session store.

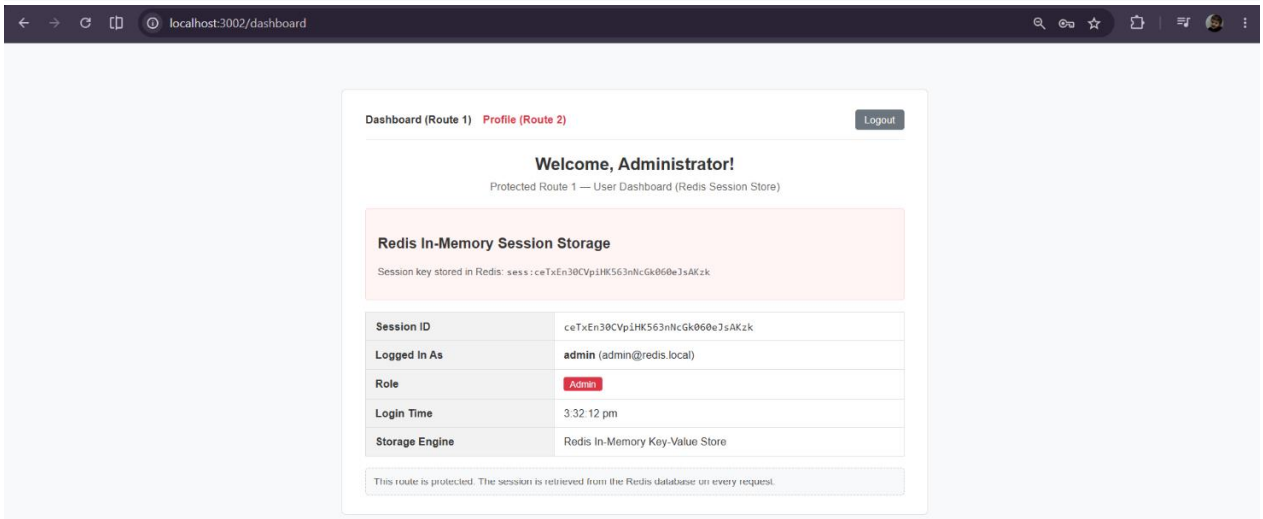

localhost:3001/login?success=Logged+out+successfully
User Login
Express application with File Session Store
Logged out successfully
Username
student
Password
Sign In
Default Credentials for Testing:
Username: admin | Password: admin123
Username: student | Password: student123

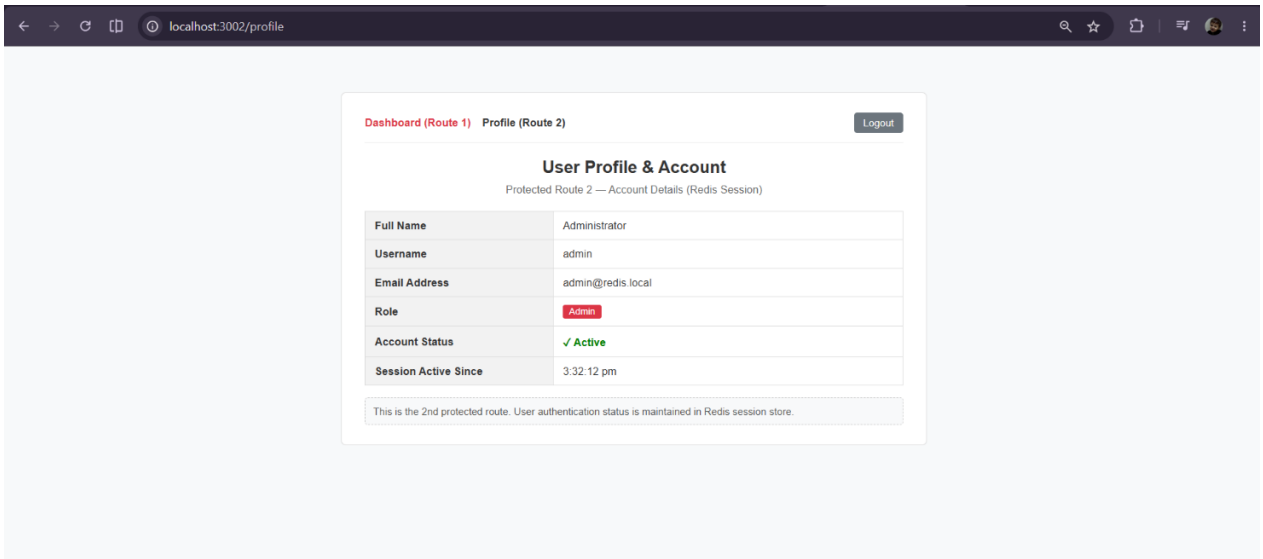

localhost:3001/dashboard
Dashboard (Route 1)
Profile (Route 2)
Logout
Welcome, John Doe!
Protected Route 1 — User Dashboard
Session Information (Stored on File System)
Your session is persisted in the server's file session store directory: ./sessions/
Session ID
tpM2-vkDYmgiy-kDomLR2gDK3vpen-wh
Session File
tpM2-vkDYmgiy-kDomLR2gDK3vpen-wh.json
Logged In As
student (student@example.com)
Role
Student
Login Timestamp
3:28:29 pm
Session Store Type
File Session Store (JSON files on disk)
This route is protected. If you logout or delete your session cookie, visiting /dashboard will automatically redirect to the login page.

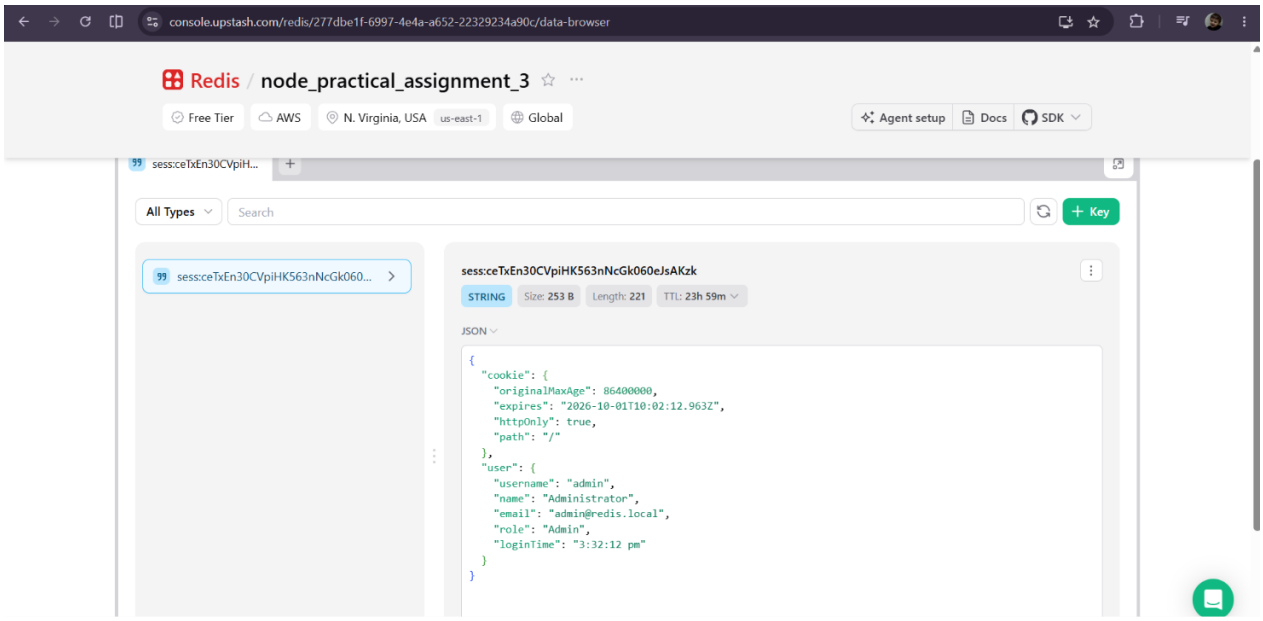

localhost:3001/profile
Dashboard (Route 1)
Profile (Route 2)
Logout
User Profile & Account
Protected Route 2 — Account Settings & Details
Full Name
John Doe
Username
student
Email Address
student@example.com
Role
Student
Account Status
✓ Active
Session Active Since
3:28:29 pm
This is the 2nd protected route. Access is verified using session authentication stored via the file session store.
Explorer
Assignment-3
Q1
app.js
package-lock.json
package.json
Q2
node_modules
public
sessions
views
app.js
package-lock.json
package.json
Q3
Preview README.md
uqN9_ephkVs8UoPzvXgX2kg6MOwObZDp.json
Problems
Output
Debug Console
Terminal
Ports
PS D:\Sem_7\701-fullstack\Assignment-3> cd .\Q2\
PS D:\Sem_7\701-fullstack\Assignment-3\Q2> node app
Server is running at http://localhost:3001
node client
node server
node Q1
node Q2

Q3

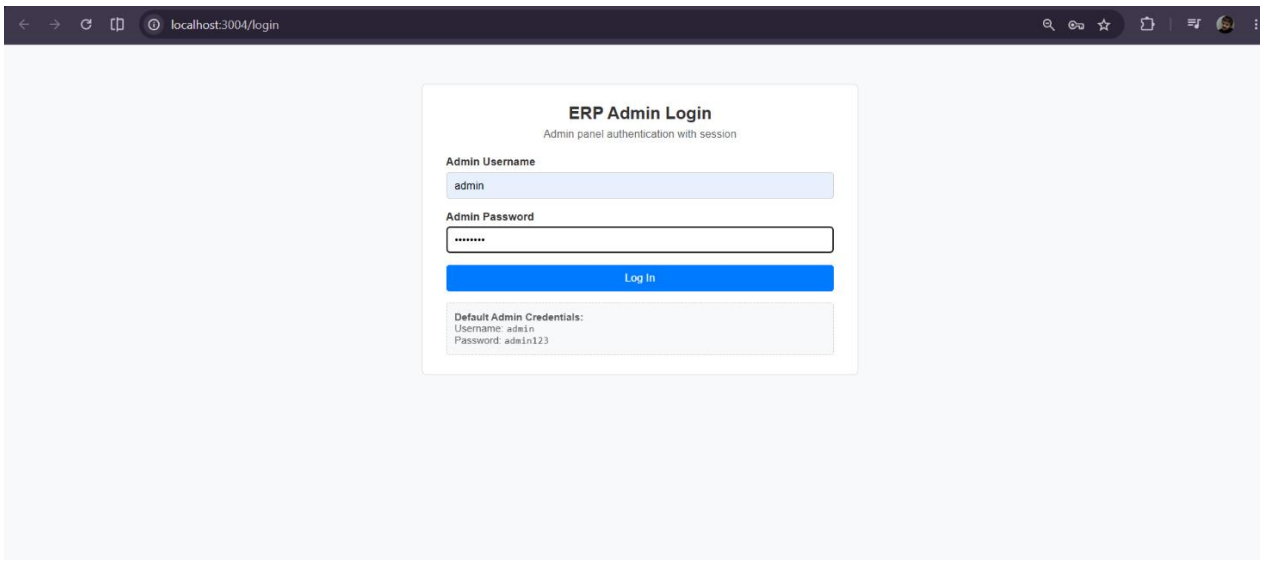

localhost:3002/login
User Login
Express application with Redis Session Store
Username
admin
Password
Sign In with Redis
Default Credentials for Testing:
Username: admin | Password: admin123
Username: student | Password: student123

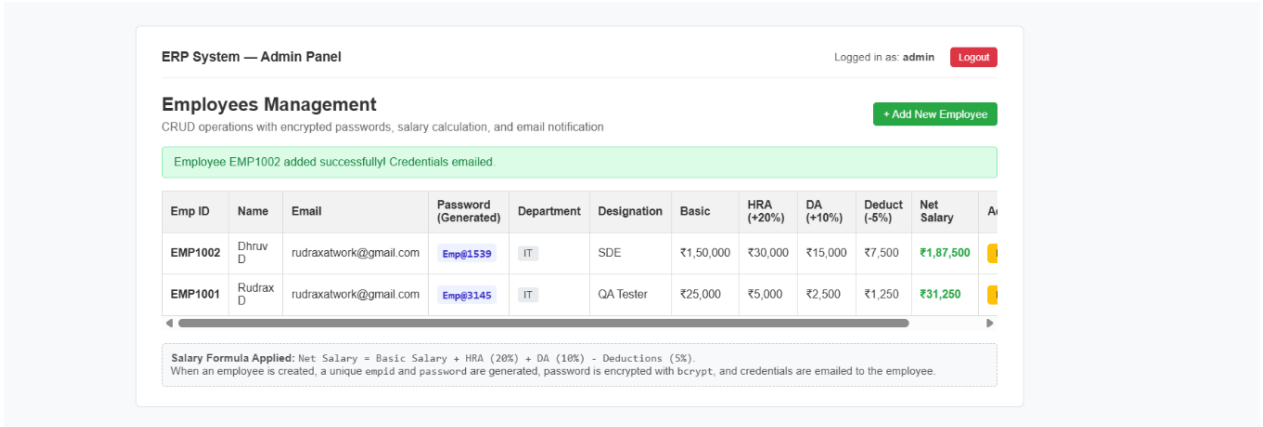

localhost:3002/dashboard
Dashboard (Route 1)
Profile (Route 2)
Logout
Welcome, Administrator!
Protected Route 1 — User Dashboard (Redis Session Store)
Redis In-Memory Session Storage
Session key stored in Redis: sess:ceTxEn30CVpiHK563nNcGk060eJsAKzk
Session ID
ceTxEn30CVpiHK563nNcGk060eJsAKzk
Logged In As
admin (admin@redis.local)
Role
Admin
Login Time
3:32:12 pm
Storage Engine
Redis In-Memory Key-Value Store
This route is protected. The session is retrieved from the Redis database on every request.

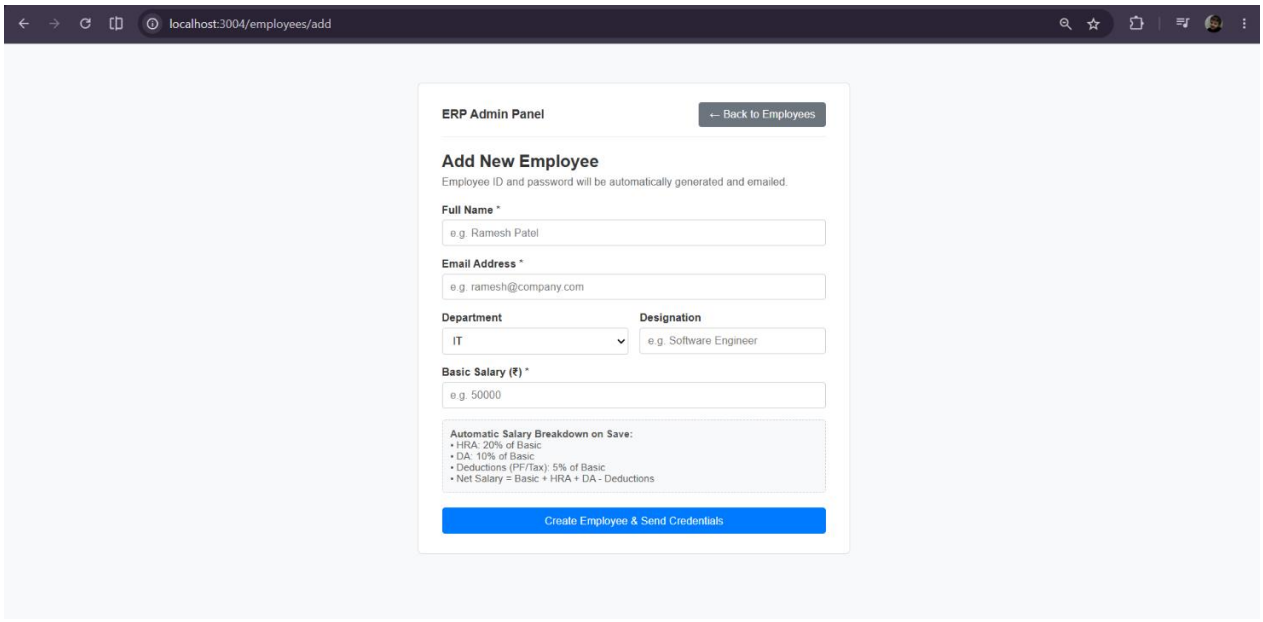

localhost:3002/profile
Dashboard (Route 1)
Profile (Route 2)
Logout
User Profile & Account
Protected Route 2 — Account Details (Redis Session)
Full Name
Administrator
Username
admin
Email Address
admin@redis.local
Role
Admin
Account Status
✓ Active
Session Active Since
3:32:12 pm
This is the 2nd protected route. User authentication status is maintained in Redis session store.

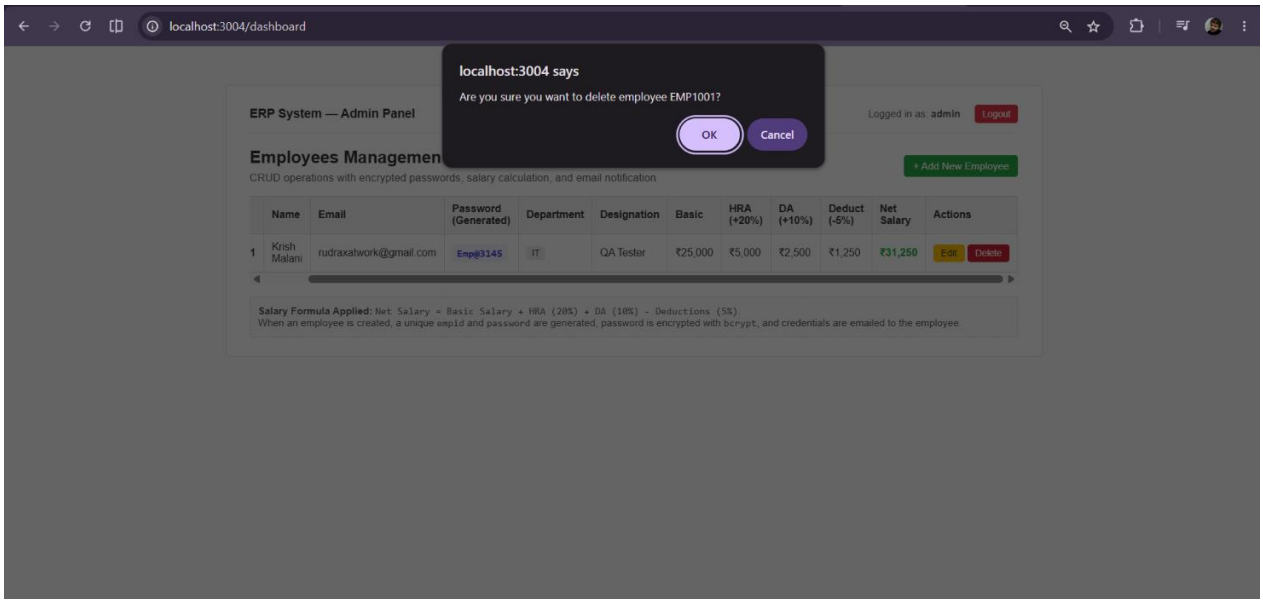

console.upstash.com/redis/277dbe1f-6997-4e4a-a652-22329234a90c/data-browser
Redis / node_practical_assignment_3
Free Tier
AWS
N. Virginia, USA us-east-1
Global
Agent setup
Docs
SDK
All Types
Search
+ Key
sess:ceTxEn30CVpiHK563nNcGk060eJsAKzk
STRING
Size: 253 B
Length: 221
TTL: 23h 59m
JSON
{
"cookie": {
"originalMaxAge": 86400000,
"expires": "2026-10-01T10:02:12.963Z",
"httpOnly": true,
"path": "/"
},
"user": {
"username": "admin",
"name": "Administrator",
"email": "admin@redis.local",
"role": "Admin",
"loginTime": "3:32:12 pm"
}
}

Q4

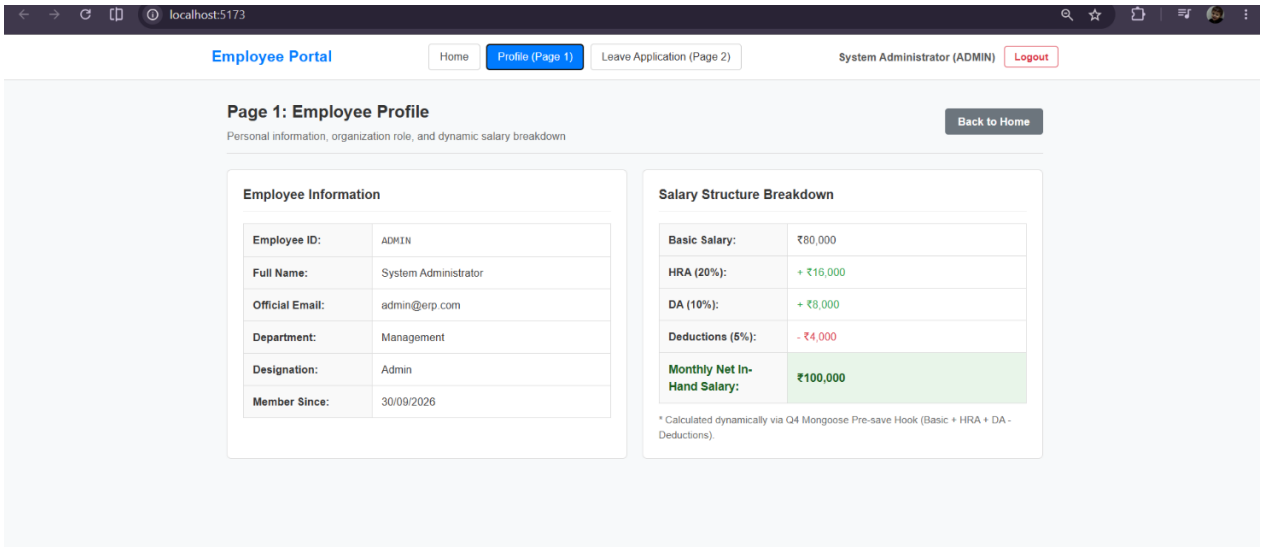

localhost:3004/login
ERP Admin Login
Admin panel authentication with session
Admin Username
admin
Admin Password
Log In
Default Admin Credentials:
Username: admin
Password: admin123

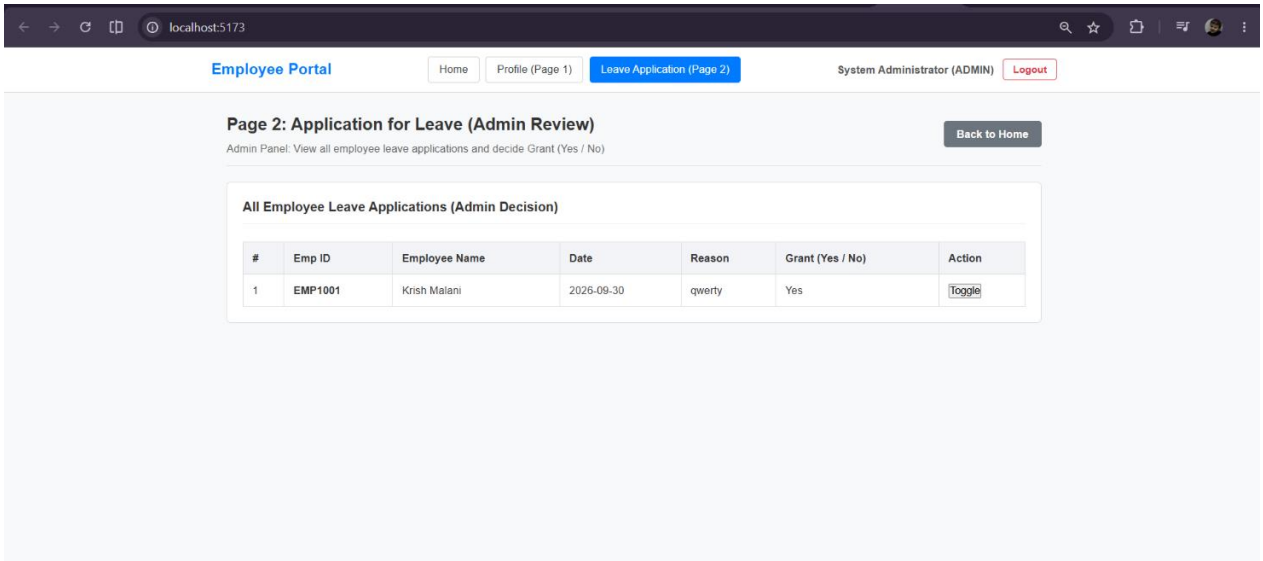

ERP System — Admin Panel
Logged in as: admin
Logout
Employees Management
CRUD operations with encrypted passwords, salary calculation, and email notification
+ Add New Employee
Employee EMP1002 added successfully! Credentials emailed.
Emp ID
Name
Email
Password (Generated)
Department
Designation
Basic
HRA (+20%)
DA (+10%)
Deduct (-5%)
Net Salary
EMP1002
Dhruv D
rudraxatwork@gmail.com
Emp@1539
IT
SDE
₹1,50,000
₹30,000
₹15,000
₹7,500
₹1,87,500
EMP1001
Rudrax D
rudraxatwork@gmail.com
Emp@3145
IT
QA Tester
₹25,000
₹5,000
₹2,500
₹1,250
₹31,250
Salary Formula Applied: Net Salary = Basic Salary + HRA (20%) + DA (10%) - Deductions (5%).
When an employee is created, a unique empid and password are generated, password is encrypted with bcrypt, and credentials are emailed to the employee.

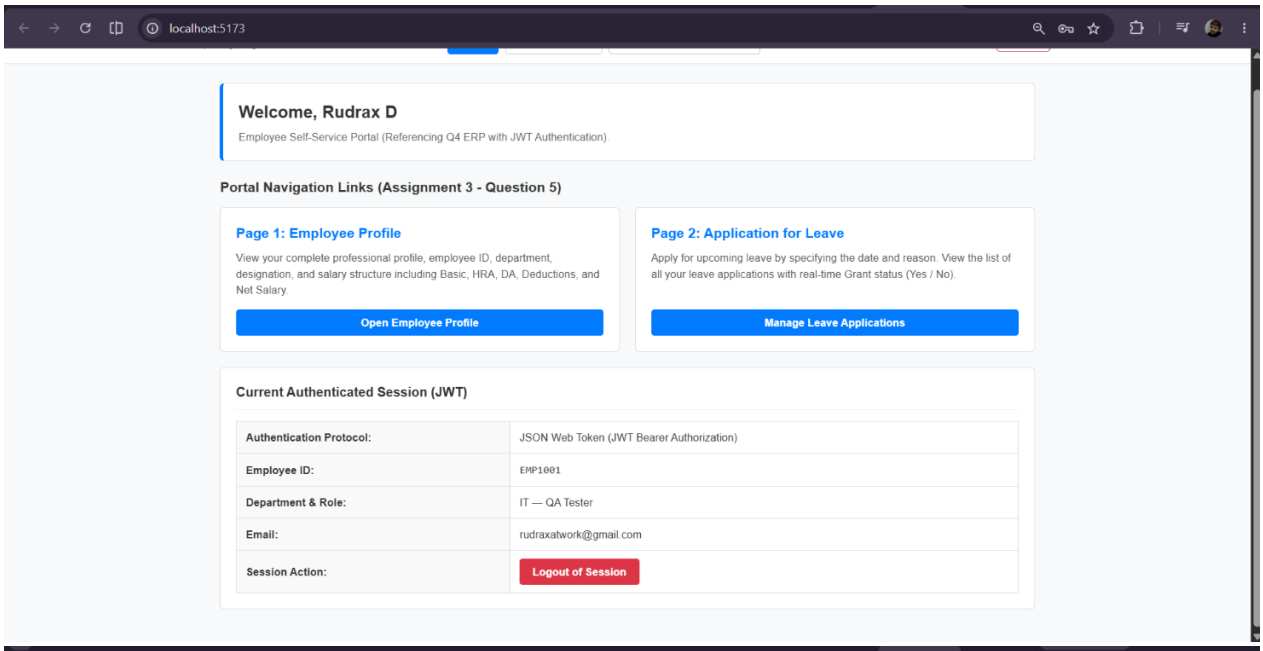

localhost:3004/employees/add
ERP Admin Panel
← Back to Employees
Add New Employee
Employee ID and password will be automatically generated and emailed.
Full Name *
e.g. Ramesh Patel
Email Address *
e.g. ramesh@company.com
Department
IT
Designation
e.g. Software Engineer
Basic Salary (₹) *
e.g. 50000
Automatic Salary Breakdown on Save:
• HRA: 20% of Basic
• DA: 10% of Basic
• Deductions (PF/Tax): 5% of Basic
• Net Salary = Basic + HRA + DA - Deductions
Create Employee & Send Credentials

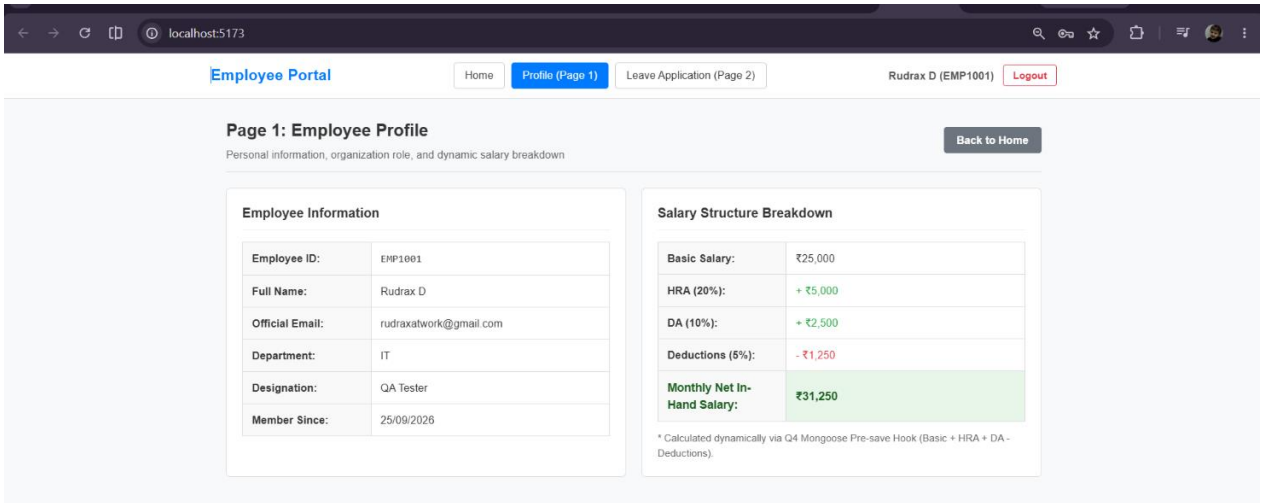

localhost:3004/dashboard
localhost:3004 says
Are you sure you want to delete employee EMP1001?
OK
Cancel
ERP System — Admin Panel
Logged in as: admin
Logout
Employees Management
CRUD operations with encrypted passwords, salary calculation, and email notification
+ Add New Employee
Name
Email
Password (Generated)
Department
Designation
Basic
HRA (+20%)
DA (+10%)
Deduct (-5%)
Net Salary
Actions
1
Krish Malani
rudraxatwork@gmail.com
Emp@3145
IT
QA Tester
₹25,000
₹5,000
₹2,500
₹1,250
₹31,250
Edit
Delete
Salary Formula Applied: Net Salary = Basic Salary + HRA (20%) + DA (10%) - Deductions (5%)
When an employee is created, a unique empId and password are generated, password is encrypted with bcrypt, and credentials are emailed to the employee.

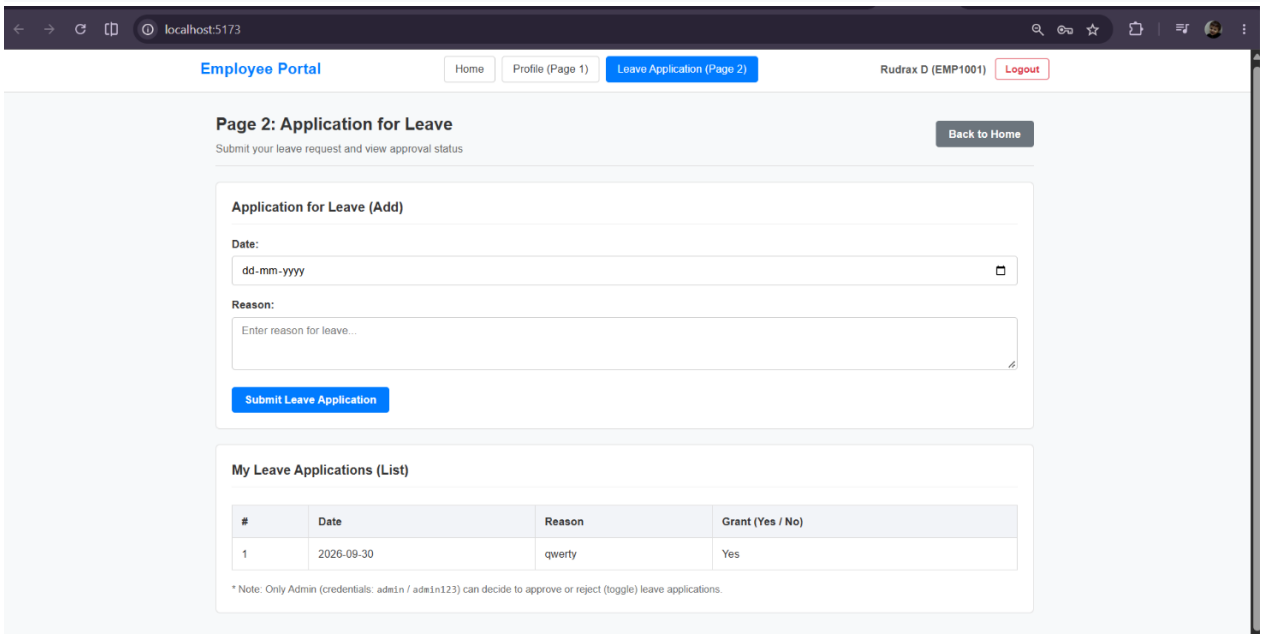
Q5


localhost:5173
Employee Portal
Home
Profile (Page 1)
Leave Application (Page 2)
System Administrator (ADMIN)
Logout
Page 1: Employee Profile
Personal information, organization role, and dynamic salary breakdown
Back to Home
Employee Information
Employee ID: ADMIN
Full Name: System Administrator
Official Email: admin@erp.com
Department: Management
Designation: Admin
Member Since: 30/09/2026
Salary Structure Breakdown
Basic Salary: ₹80,000
HRA (20%): + ₹16,000
DA (10%): + ₹8,000
Deductions (5%): - ₹4,000
Monthly Net In-Hand Salary: ₹100,000
* Calculated dynamically via Q4 Mongoose Pre-save Hook (Basic + HRA + DA - Deductions).


localhost:5173
Employee Portal
Home
Profile (Page 1)
Leave Application (Page 2)
System Administrator (ADMIN)
Logout
Page 2: Application for Leave (Admin Review)
Admin Panel: View all employee leave applications and decide Grant (Yes / No)
Back to Home
All Employee Leave Applications (Admin Decision)
Emp ID Employee Name Date Reason Grant (Yes / No) Action
1 EMP1001 Krish Malani 2026-09-30 qwerty Yes Toggle


localhost:5173
Welcome, Rudrax D
Employee Self-Service Portal (Referencing Q4 ERP with JWT Authentication).
Portal Navigation Links (Assignment 3 - Question 5)
Page 1: Employee Profile
View your complete professional profile, employee ID, department, designation, and salary structure including Basic, HRA, DA, Deductions, and Net Salary.
Open Employee Profile
Page 2: Application for Leave
Apply for upcoming leave by specifying the date and reason. View the list of all your leave applications with real-time Grant status (Yes / No).
Manage Leave Applications
Current Authenticated Session (JWT)
Authentication Protocol: JSON Web Token (JWT Bearer Authorization)
Employee ID: EMP1001
Department & Role: IT — QA Tester
Email: rudraxatwork@gmail.com
Session Action: Logout of Session


localhost:5173
Employee Portal
Home
Profile (Page 1)
Leave Application (Page 2)
Rudrax D (EMP1001)
Logout
Page 1: Employee Profile
Personal information, organization role, and dynamic salary breakdown
Back to Home
Employee Information
Employee ID: EMP1001
Full Name: Rudrax D
Official Email: rudraxatwork@gmail.com
Department: IT
Designation: QA Tester
Member Since: 25/09/2026
Salary Structure Breakdown
Basic Salary: ₹25,000
HRA (20%): + ₹5,000
DA (10%): + ₹2,500
Deductions (5%): - ₹1,250
Monthly Net In-Hand Salary: ₹31,250
* Calculated dynamically via Q4 Mongoose Pre-save Hook (Basic + HRA + DA - Deductions).

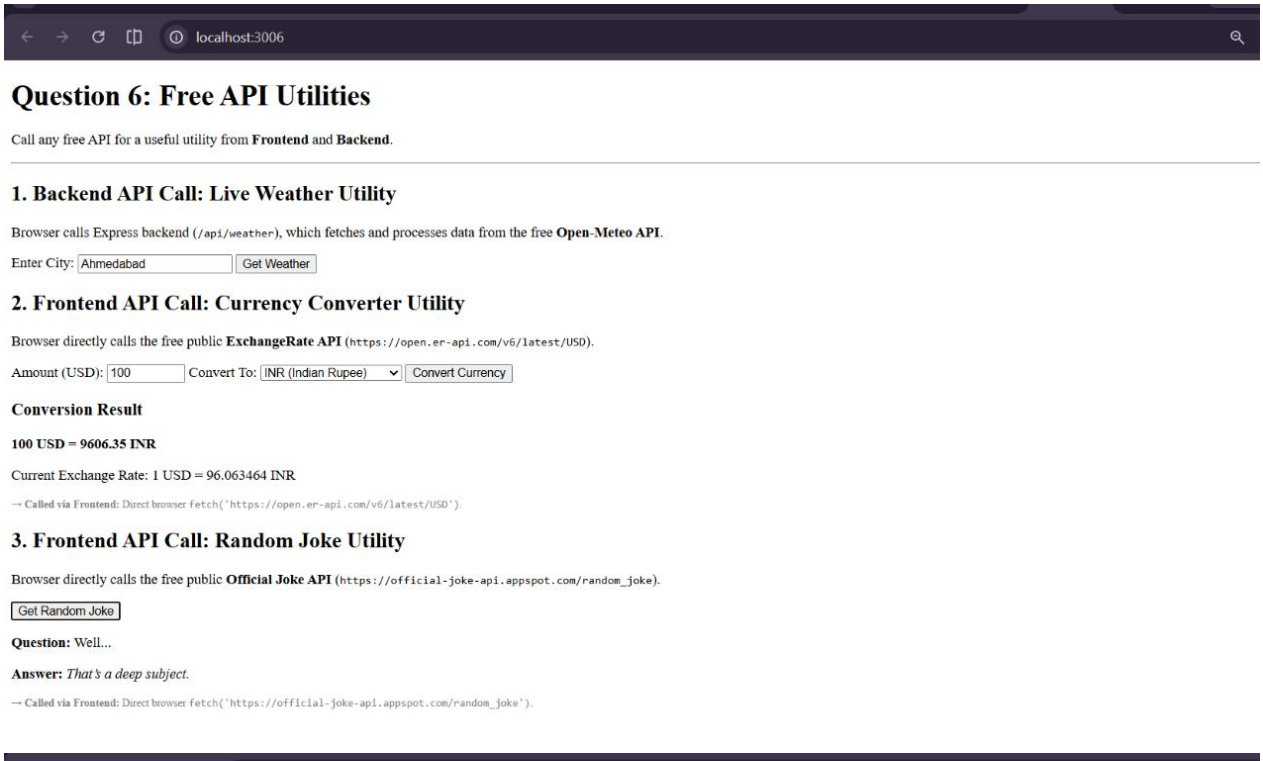

localhost:5173
Employee Portal
Home
Profile (Page 1)
Leave Application (Page 2)
Rudrax D (EMP1001)
Logout
Page 2: Application for Leave
Submit your leave request and view approval status
Back to Home
Application for Leave (Add)
Date:
dd-mm-yyyy
Reason:
Enter reason for leave...
Submit Leave Application
My Leave Applications (List)
Date Reason Grant (Yes / No)
1 2026-09-30 qwerty Yes
* Note: Only Admin (credentials: admin / admin123) can decide to approve or reject (toggle) leave applications.

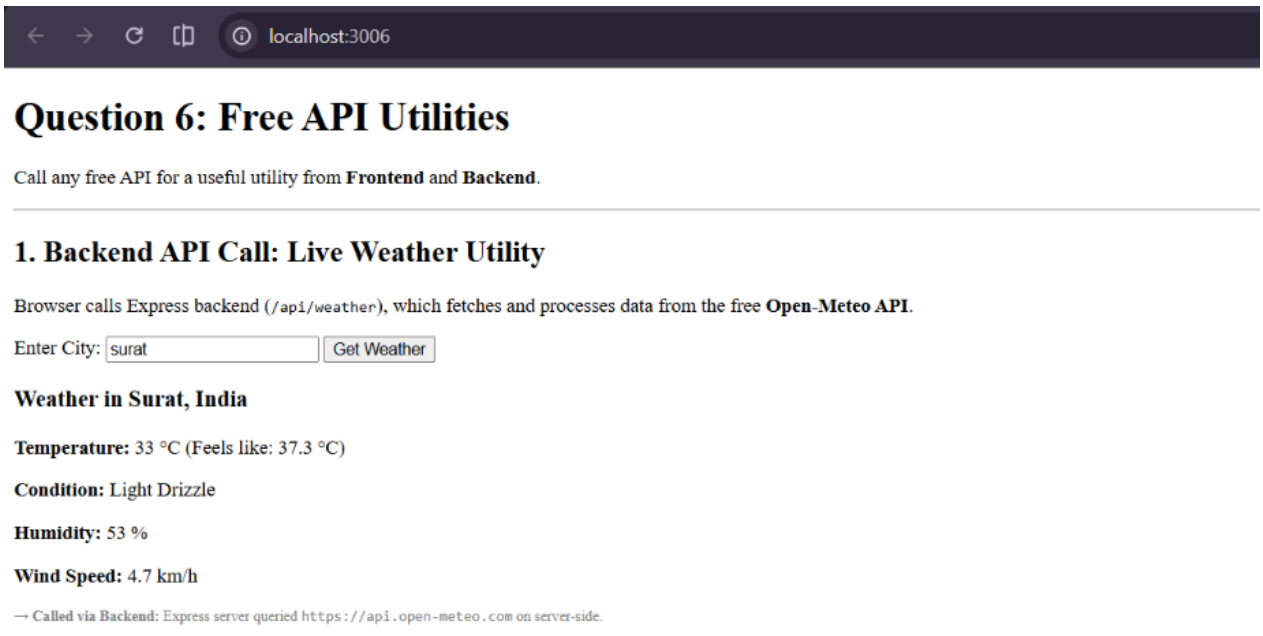
Q6

```
PS D:\Sem_7\701-fullstack\Assignment-3\Q6> npm start

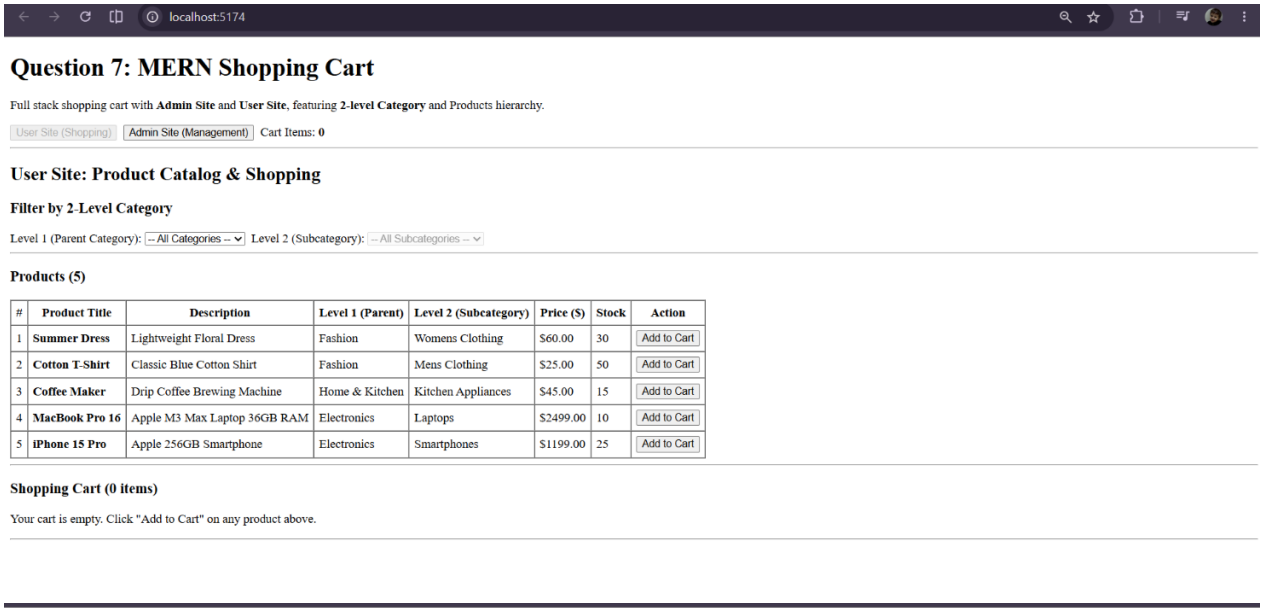
> q6-free-api-utility@1.0.0 start
> node server.js

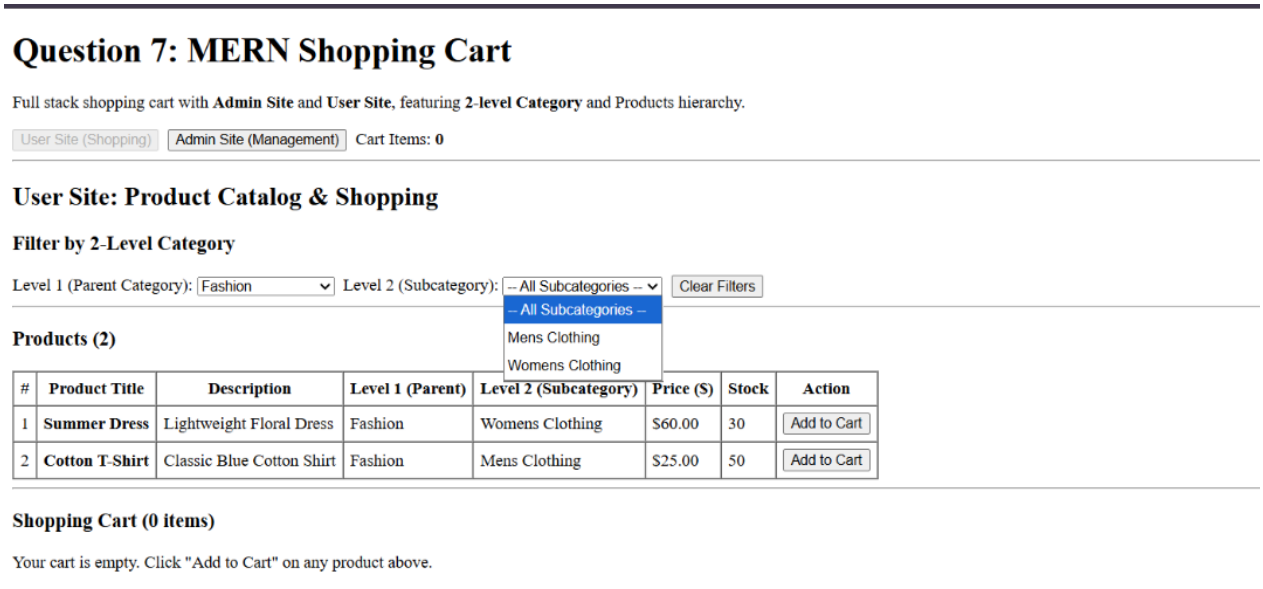
============================================================
 Q6 Utility Service running at http://localhost:3006
 Backend API Endpoint: http://localhost:3006/api/weather?city=Mumbai
 Frontend UI available at: http://localhost:3006
============================================================
```





Q7


localhost:5174
Question 7: MERN Shopping Cart
Full stack shopping cart with Admin Site and User Site, featuring 2-level Category and Products hierarchy.
User Site (Shopping)
Admin Site (Management)
Cart Items: 0
User Site: Product Catalog & Shopping
Filter by 2-Level Category
Level 1 (Parent Category): -- All Categories --
Level 2 (Subcategory): -- All Subcategories --
Products (5)
Product Title Description Level 1 (Parent) Level 2 (Subcategory) Price ($) Stock Action
1 Summer Dress Lightweight Floral Dress Fashion Womens Clothing $60.00 30 Add to Cart
2 Cotton T-Shirt Classic Blue Cotton Shirt Fashion Mens Clothing $25.00 50 Add to Cart
3 Coffee Maker Drip Coffee Brewing Machine Home & Kitchen Kitchen Appliances $45.00 15 Add to Cart
4 MacBook Pro 16 Apple M3 Max Laptop 36GB RAM Electronics Laptops $2499.00 10 Add to Cart
5 iPhone 15 Pro Apple 256GB Smartphone Electronics Smartphones $1199.00 25 Add to Cart
Shopping Cart (0 items)
Your cart is empty. Click "Add to Cart" on any product above.


Question 7: MERN Shopping Cart
Full stack shopping cart with Admin Site and User Site, featuring 2-level Category and Products hierarchy.
User Site (Shopping)
Admin Site (Management)
Cart Items: 0
User Site: Product Catalog & Shopping
Filter by 2-Level Category
Level 1 (Parent Category): Fashion
Level 2 (Subcategory): -- All Subcategories --
Clear Filters
-- All Subcategories --
Mens Clothing
Womens Clothing
Products (2)
Product Title Description Level 1 (Parent) Level 2 (Subcategory) Price ($) Stock Action
1 Summer Dress Lightweight Floral Dress Fashion Womens Clothing $60.00 30 Add to Cart
2 Cotton T-Shirt Classic Blue Cotton Shirt Fashion Mens Clothing $25.00 50 Add to Cart
Shopping Cart (0 items)
Your cart is empty. Click "Add to Cart" on any product above.

Question 7: MERN Shopping Cart

Full stack shopping cart with **Admin Site** and **User Site**, featuring **2-level Category** and Products hierarchy.

User Site (Shopping) | Admin Site (Management) | Cart Items: **1**

User Site: Product Catalog & Shopping

Filter by 2-Level Category

Level 1 (Parent Category): Fashion | Level 2 (Subcategory): -- All Subcategories -- | Clear Filters

Products (2)

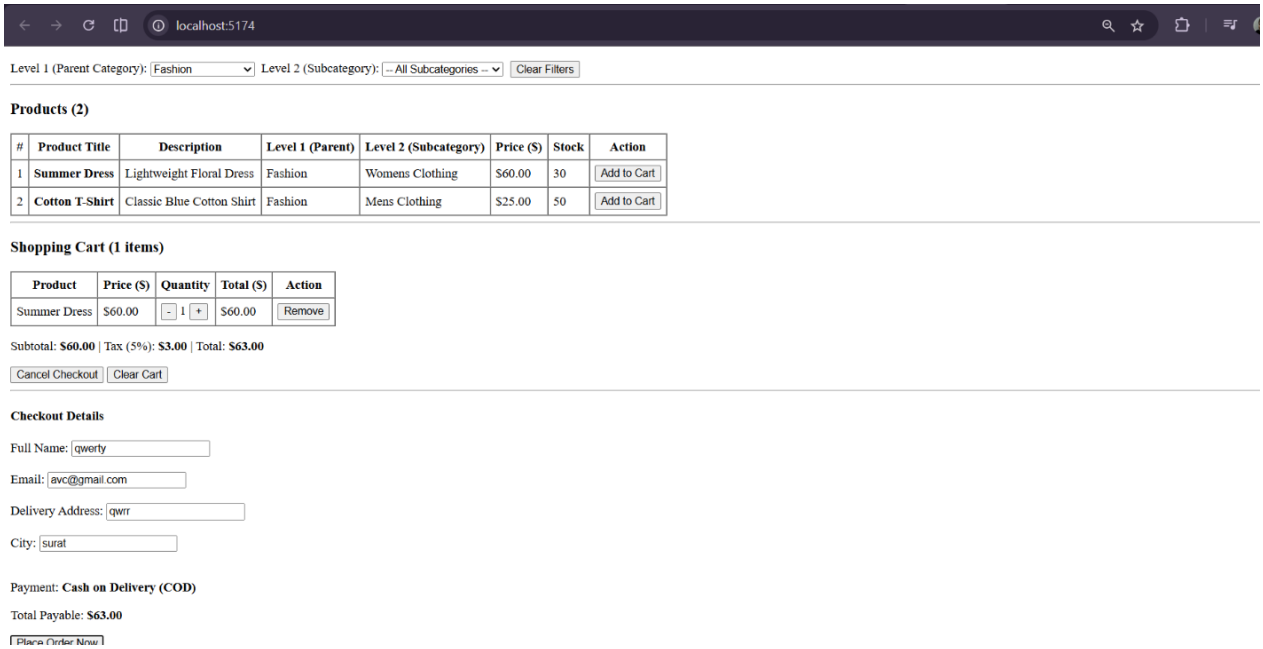
#	Product Title	Description	Level 1 (Parent)	Level 2 (Subcategory)	Price ($)	Stock	Action
1	**Summer Dress**	Lightweight Floral Dress	Fashion	Womens Clothing	$60.00	30	Add to Cart
2	**Cotton T-Shirt**	Classic Blue Cotton Shirt	Fashion	Mens Clothing	$25.00	50	Add to Cart

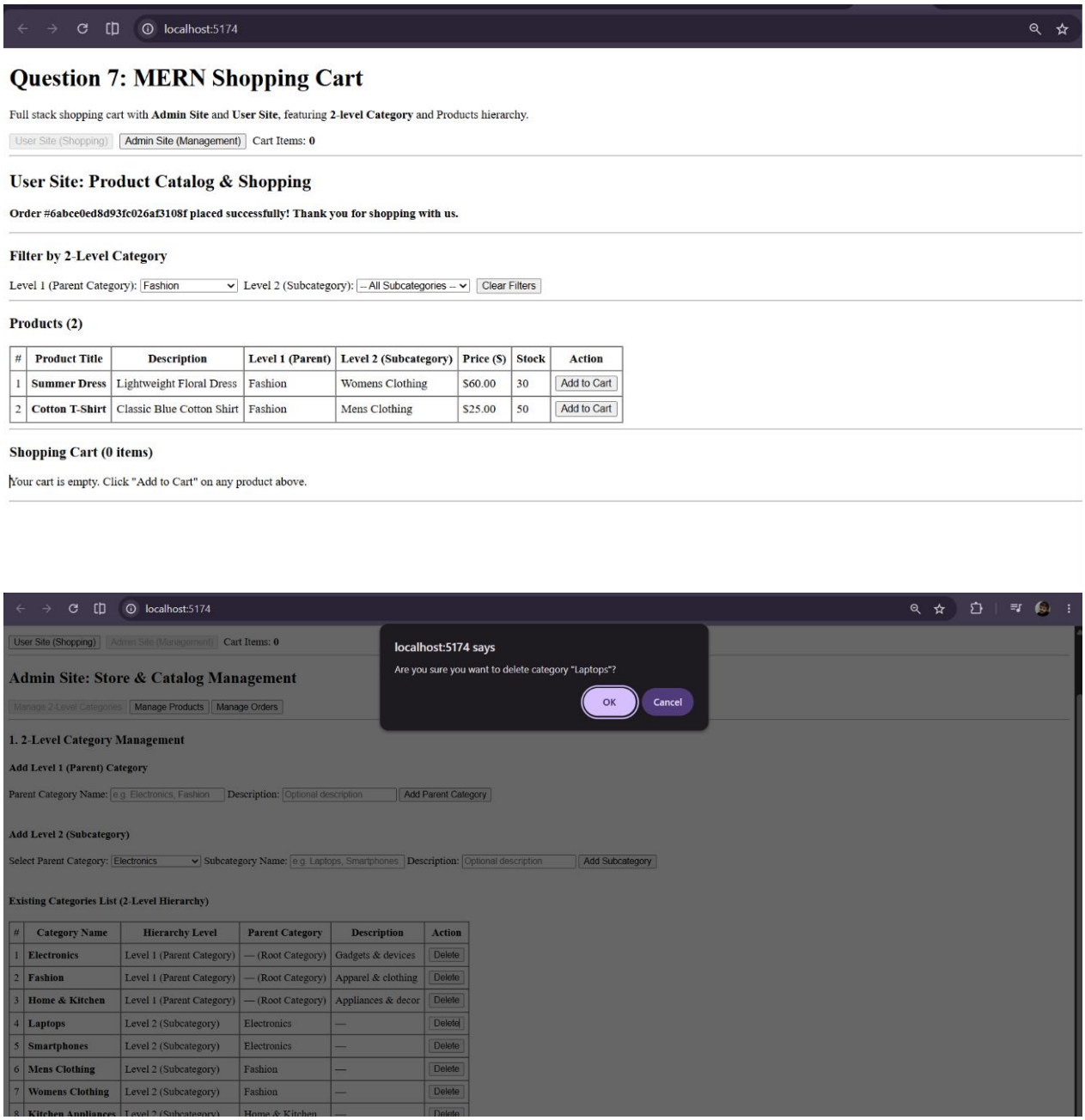
Shopping Cart (1 items)

Product	Price ($)	Quantity	Total ($)	Action
Summer Dress	$60.00	- 1 +	$60.00	Remove

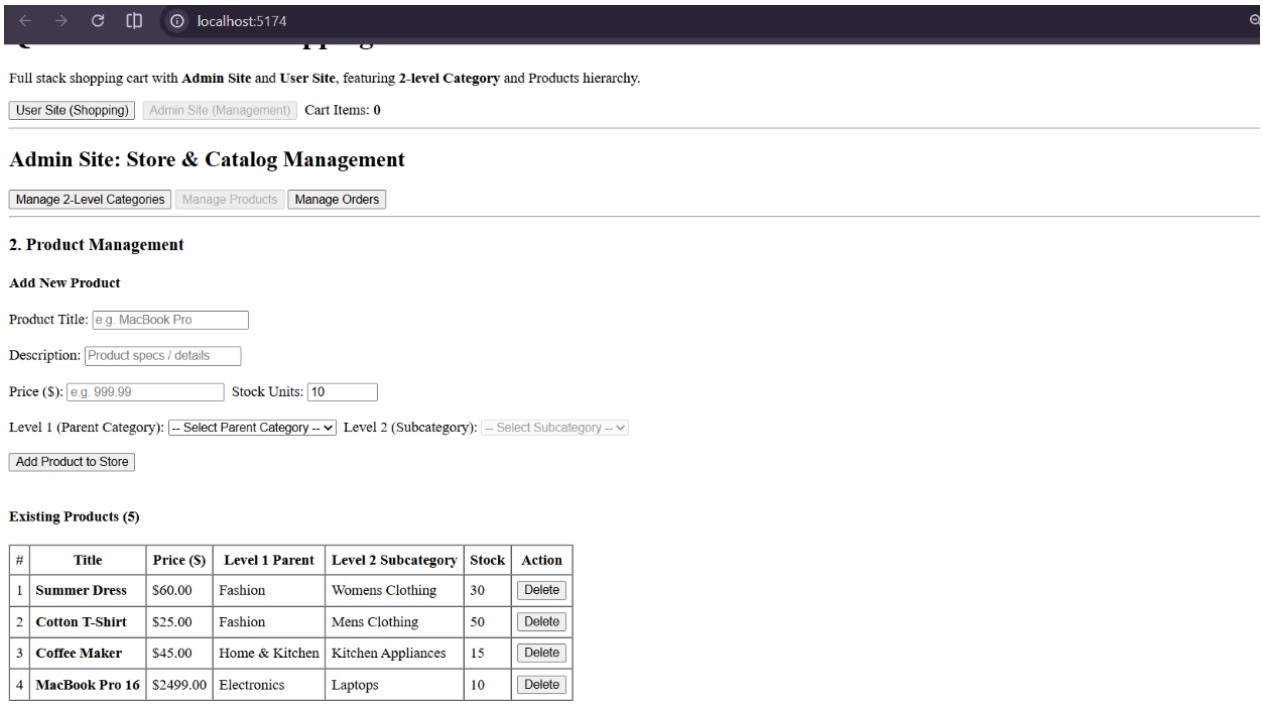
Subtotal: **$60.00** | Tax (5%): **$3.00** | Total: **$63.00**

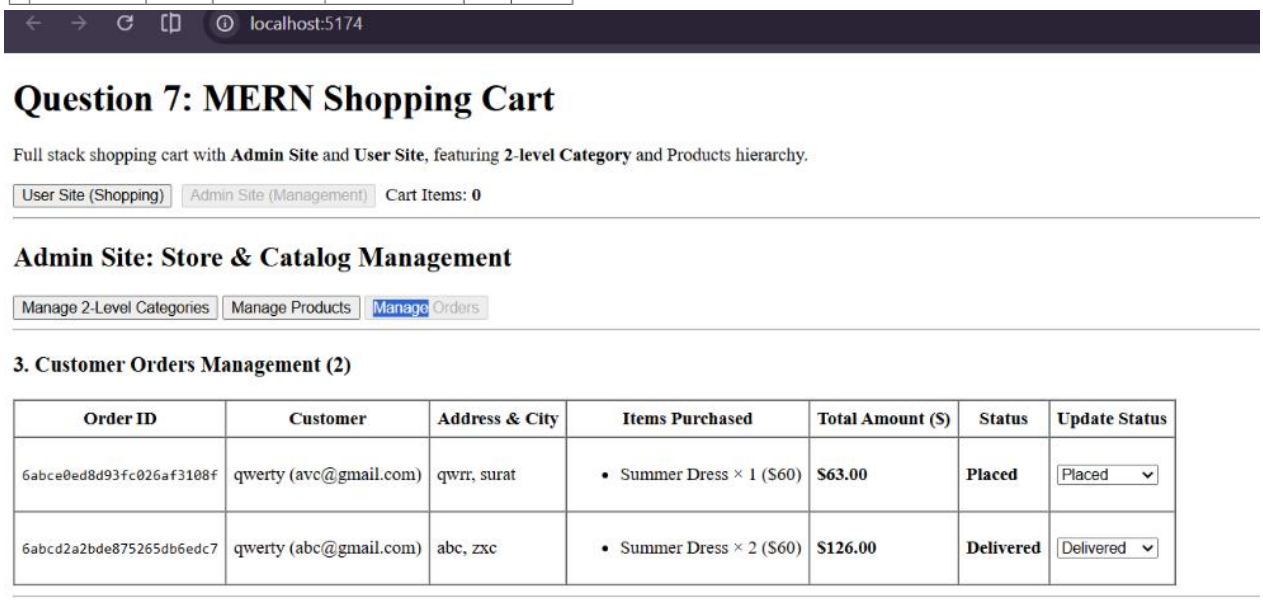
Proceed to Checkout | Clear Cart




localhost:5174
Question 7: MERN Shopping Cart
Full stack shopping cart with Admin Site and User Site, featuring 2-level Category and Products hierarchy.
User Site (Shopping)
Admin Site (Management)
Cart Items: 0
User Site: Product Catalog & Shopping
Order #6abce0ed8d93fc026af3108f placed successfully! Thank you for shopping with us.
Filter by 2-Level Category
Level 1 (Parent Category): Fashion
Level 2 (Subcategory): -- All Subcategories --
Clear Filters
Products (2)
Product Title Description Level 1 (Parent) Level 2 (Subcategory) Price ($) Stock Action
1 Summer Dress Lightweight Floral Dress Fashion Womens Clothing $60.00 30 Add to Cart
2 Cotton T-Shirt Classic Blue Cotton Shirt Fashion Mens Clothing $25.00 50 Add to Cart
Shopping Cart (0 items)
Your cart is empty. Click "Add to Cart" on any product above.

localhost:5174
localhost:5174 says
Are you sure you want to delete category "Laptops"?
OK
Cancel
User Site (Shopping)
Admin Site (Management)
Cart Items: 0
Admin Site: Store & Catalog Management
Manage 2-Level Categories
Manage Products
Manage Orders
1. 2-Level Category Management
Add Level 1 (Parent) Category
Parent Category Name: e.g. Electronics, Fashion
Description: Optional description
Add Parent Category
Add Level 2 (Subcategory)
Select Parent Category: Electronics
Subcategory Name: e.g. Laptops, Smartphones
Description: Optional description
Add Subcategory
Existing Categories List (2-Level Hierarchy)
Category Name Hierarchy Level Parent Category Description Action
1 Electronics Level 1 (Parent Category) — (Root Category) Gadgets & devices Delete
2 Fashion Level 1 (Parent Category) — (Root Category) Apparel & clothing Delete
3 Home & Kitchen Level 1 (Parent Category) — (Root Category) Appliances & decor Delete
4 Laptops Level 2 (Subcategory) Electronics — Delete
5 Smartphones Level 2 (Subcategory) Electronics — Delete
6 Mens Clothing Level 2 (Subcategory) Fashion — Delete
7 Womens Clothing Level 2 (Subcategory) Fashion — Delete
8 Kitchen Appliances Level 2 (Subcategory) Home & Kitchen — Delete


localhost:5174
Full stack shopping cart with Admin Site and User Site, featuring 2-level Category and Products hierarchy.
User Site (Shopping)
Admin Site (Management)
Cart Items: 0
Admin Site: Store & Catalog Management
Manage 2-Level Categories
Manage Products
Manage Orders
2. Product Management
Add New Product
Product Title:
e.g. MacBook Pro
Description:
Product specs / details
Price ($):
e.g. 999.99
Stock Units:
10
Level 1 (Parent Category):
-- Select Parent Category --
Level 2 (Subcategory):
-- Select Subcategory --
Add Product to Store
Existing Products (5)
Title Price ($) Level 1 Parent Level 2 Subcategory Stock Action
1 Summer Dress $60.00 Fashion Womens Clothing 30 Delete
2 Cotton T-Shirt $25.00 Fashion Mens Clothing 50 Delete
3 Coffee Maker $45.00 Home & Kitchen Kitchen Appliances 15 Delete
4 MacBook Pro 16 $2499.00 Electronics Laptops 10 Delete


localhost:5174
Question 7: MERN Shopping Cart
Full stack shopping cart with Admin Site and User Site, featuring 2-level Category and Products hierarchy.
User Site (Shopping)
Admin Site (Management)
Cart Items: 0
Admin Site: Store & Catalog Management
Manage 2-Level Categories
Manage Products
Manage Orders
3. Customer Orders Management (2)
Order ID Customer Address & City Items Purchased Total Amount ($) Status Update Status
6abce0ed8d93fc026af3108f qwerty (avc@gmail.com) qwrr, surat • Summer Dress × 1 ($60) $63.00 Placed Placed
6abcd2a2bde875265db6edc7 qwerty (abc@gmail.com) abc, zxc • Summer Dress × 2 ($60) $126.00 Delivered Delivered

Q8



Question 8: Student CRUD Application

Technology Stack: **Express.js + Sequelize ORM + React.js (Vite)** (No CSS).

Add New Student

Roll Number: e.g. CS104

Full Name: e.g. Ananya Desai

Email Address: e.g. ananya@college.edu

Course / Department: Computer Engineering

Semester: 1

CPI / Grade: 8.0

Add Student

Student Records List (3)

Search Students: Search by Roll No, Name, o

#	Roll No	Full Name	Email	Course	Semester	CPI	Actions
1	**CS104**	Raghav Chadda	rc@gmail.com	Mechanical Engineering	5	9.00	Edit Delete
2	**CS101**	Rohan Sharma	rohan.sharma@college.edu	Computer Engineering	7	8.75	Edit Delete
3	**CS102**	Priya Patel	priya.patel@college.edu	Information Technology	5	9.20	Edit Delete

Sequelize Model: `Student (rollNo, name, email, course, semester, cpi)` • Database: SQLite (Zero Setup)



Question 8: Student CRUD Application

Technology Stack: **Express.js + Sequelize ORM + React.js (Vite)** (No CSS).

Edit Student Details

Roll Number: CS104

Full Name: Raghav Chadda

Email Address: rc@gmail.com

Course / Department: Mechanical Engineering

Semester: 5

CPI / Grade: 9

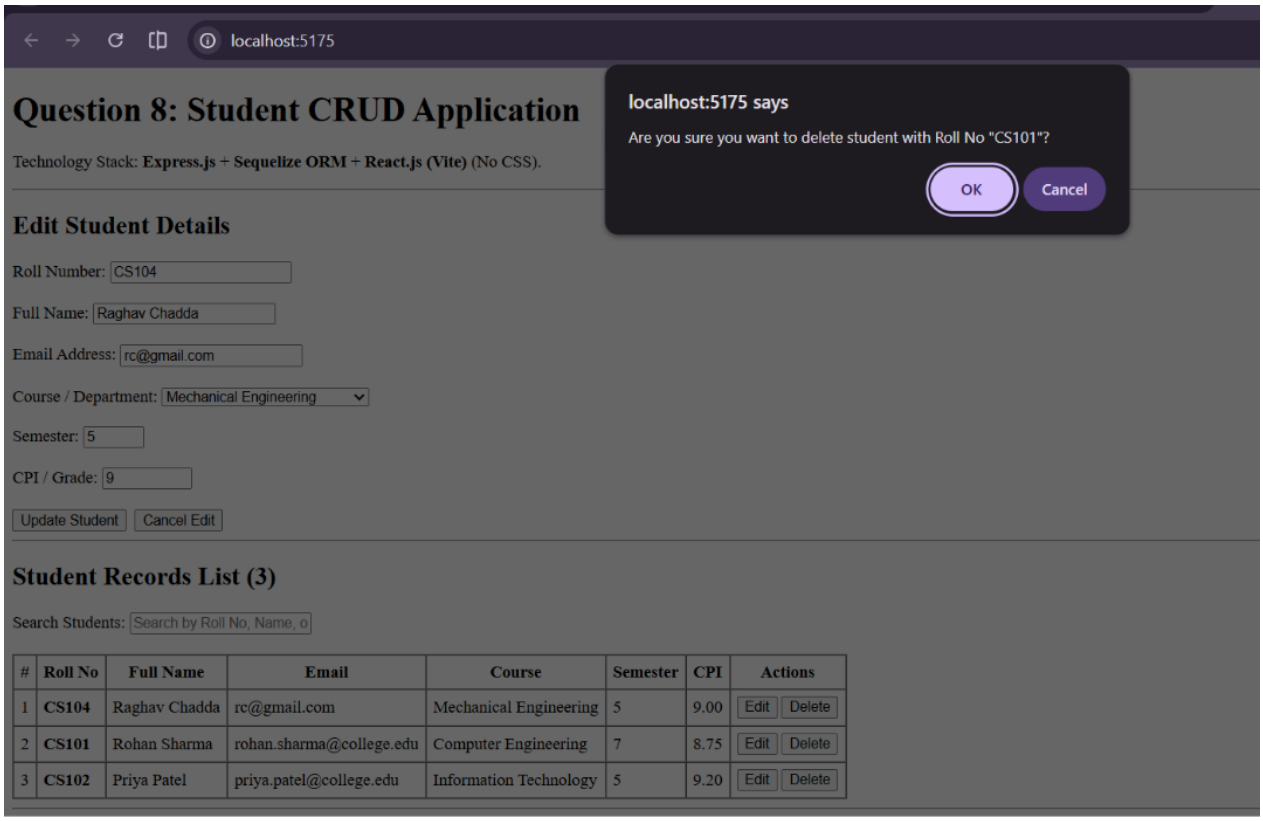
Update Student Cancel Edit

Student Records List (3)

Search Students: Search by Roll No, Name, o

#	Roll No	Full Name	Email	Course	Semester	CPI	Actions
1	**CS104**	Raghav Chadda	rc@gmail.com	Mechanical Engineering	5	9.00	Edit Delete
2	**CS101**	Rohan Sharma	rohan.sharma@college.edu	Computer Engineering	7	8.75	Edit Delete
3	**CS102**	Priya Patel	priya.patel@college.edu	Information Technology	5	9.20	Edit Delete

Sequelize Model: `Student (rollNo, name, email, course, semester, cpi)` • Database: SQLite (Zero Setup)


localhost:5175
Question 8: Student CRUD Application
Technology Stack: Express.js + Sequelize ORM + React.js (Vite) (No CSS).
localhost:5175 says
Are you sure you want to delete student with Roll No "CS101"?
OK
Cancel
Edit Student Details
Roll Number: CS104
Full Name: Raghav Chadda
Email Address: rc@gmail.com
Course / Department: Mechanical Engineering
Semester: 5
CPI / Grade: 9
Update Student
Cancel Edit
Student Records List (3)
Search Students: Search by Roll No, Name, o
Roll No Full Name Email Course Semester CPI Actions
1 CS104 Raghav Chadda rc@gmail.com Mechanical Engineering 5 9.00 Edit Delete
2 CS101 Rohan Sharma rohan.sharma@college.edu Computer Engineering 7 8.75 Edit Delete
3 CS102 Priya Patel priya.patel@college.edu Information Technology 5 9.20 Edit Delete